Smart Mouse

User Documentation

TelePresent Games · 2026

Contents

Introduction

Thank you for checking out Smart Mouse! I hope it'll help streamline your daily work in Unity, and that you'll get as much enjoyment from it as I have 🙂

Smart Mouse is a set of lightweight tools for the Unity **Scene View**, all reachable from a few clicks or key presses. It is built to feel like part of the editor, and unobtrusive to your workflow.

This documentation covers installation, every feature, and how to add your own context-menu actions.

What it does

Smart Mouse Includes the following features:

> **Select anything under the cursor.** Hold a key and scroll through every object beneath the pointer, meshes, colliders, sprites and world-space UI alike, no matter what is drawn on top, then click to select. See the Smart Selection Overlay.

> **Place objects on surfaces.** Drag them into place with the Surface Snap Tool, or paste at the cursor, snapped and aligned to the surface.

> **Smart Right-click menu.** The Smart Context Menu lets you create objects and prefabs on surfaces, snap & align objects anywhere, place objects with physics, and much more. You can even add your own entries.

Hover to highlight and select anything under the cursor, right in the Scene View.

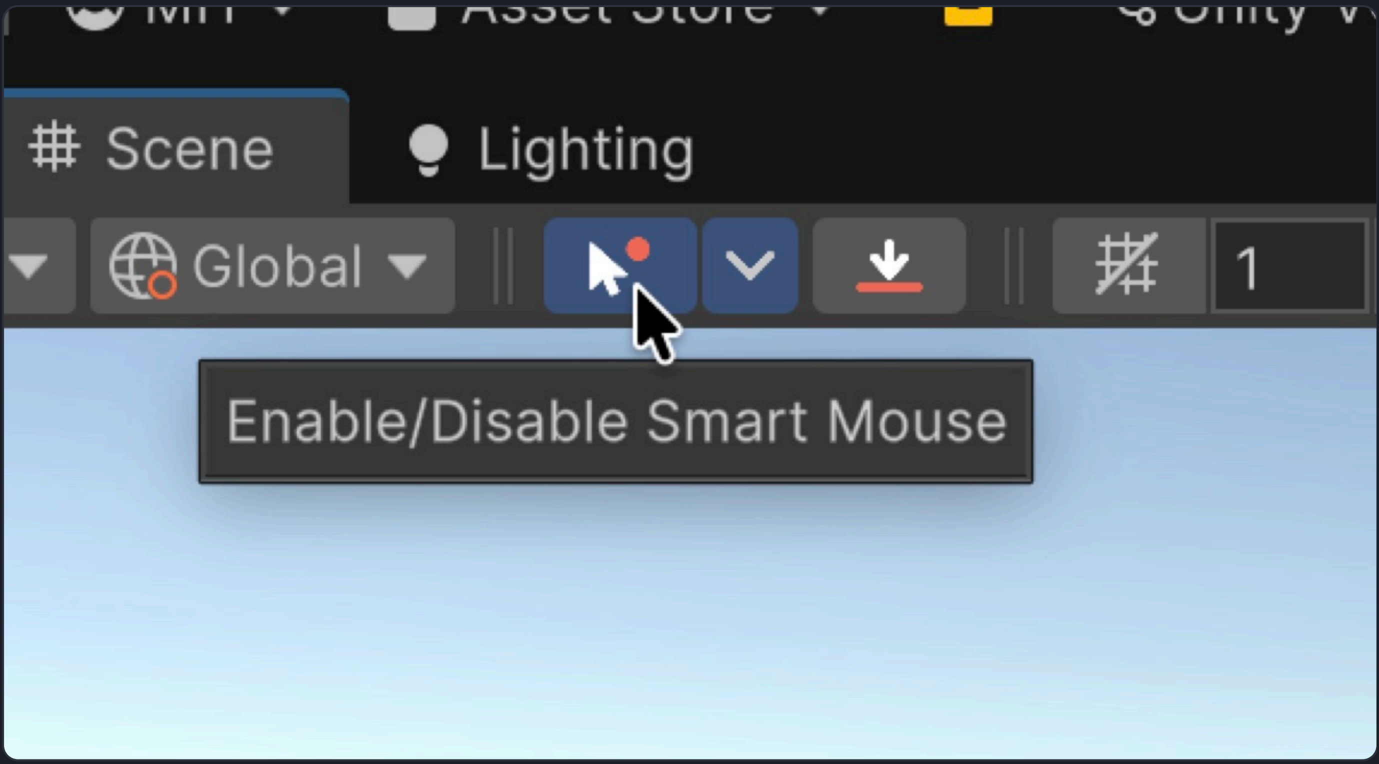
Smart Mouse works is tested for-and works with Unity 2022 LTS and newer, including Unity 6. To get going, head to Installation, or open the Demo Scene to see it in action.

Installation

Import the asset from the **Package Manager**, under *My Assets*. The files install to **Assets/TelePresent/Smart Mouse**, and that's it!

The toolbar button

After importing, you should see the **Smart Mouse button** on your Scene View toolbar. Clicking it toggles Smart Mouse on or off (it lights up blue when enabled), and the small dropdown arrow beside it opens the Smart Mouse Settings.



The Smart Mouse toggle button (blue when active) and its settings dropdown arrow.

> **Don't see the button?**
> First, try **restarting the editor**. If it still doesn't appear, open the Scene View's **overlay menu** (the overlays button in the top-right of the Scene View), enable **Smart Mouse Toolbar**, then drag it up into the toolbar to dock it.

The Tools menu

Smart Mouse also adds a **Tools ▸ TelePresent ▸ Smart Mouse** menu. From there you can enable or disable Smart Mouse, open its tool windows, activate the Surface Snap Tool, toggle the Measure mode, and open this documentation.

Requirements

> **Compatibility**
>
> Smart Mouse supports **Unity 2022 LTS and newer**, and is fully supported on **Unity 6**, across all render pipelines.
>
> If you're using Unity 2021, please let me know, and I'll see if I can get support for this too :)

With that done, head to the Smart Selection Overlay to start hovering and selecting.

GETTING STARTED

Smart Selection Overlay

The Smart Selection Overlay lets you scroll through every object under the cursor, no matter what is culled or drawn on top, and pick exactly the one you want. It is a quick way to grab an object buried inside a cluster without hunting through the Hierarchy.

How to use it

> **Hold the Overlay Key** to activate it. The default is ⌘ **(Command)** on macOS and **Ctrl** on Windows, and you can change it in Settings.

> **Hover** over the scene. The object under the cursor is highlighted, and a small label shows its name and the world position you are pointing at.

> **Scroll the mouse wheel** to step through every object stacked under the cursor, front to back.

> **Click** to select the highlighted object. **Shift-click** adds it to the selection, or removes it.

An object highlighted under the cursor, with its name and position label.

What it can detect

In Settings you choose which kinds of objects the overlay considers:

> **Meshes:** per-triangle detection that works even on objects with no collider. Covers Mesh Renderers and Skinned Mesh Renderers.

> **Colliders:** uses 2D and 3D physics colliders. Lighter than mesh detection, which helps on very heavy scenes.

> **Sprites:** world-space **Sprite Renderers** are always detected.

> **UI Elements:** hover and select world-space Canvas UI such as panels, images and buttons, highlighted with the same look as meshes. In the Settings, you can define whether to include invisible UI elements as well.

That last one is especially handy. World-space Canvas UI is awkward to click directly in the Scene View, but with Canvas UI detection on you can hover, scroll and select panels, images and buttons just like any other object.

Hovering and selecting world-space Canvas UI in the Scene View.

> **Works everywhere**
> Smart Selection works across all render pipelines and has been tested with custom shaders. It ignores scene **gizmos** such as light and camera icons. It also works in **Prefab Mode**: while editing a prefab in isolation, the overlay sees only the prefab's own objects, and it switches back to the scene the moment you exit.

Selecting Canvas UI

Canvas UI is awkward to click in the Scene View, so Smart Mouse detects it the same way it detects everything else. The **Canvas UI** setting in Settings has three states:

> **Visible only (default):** picks elements that actually draw. An element that is fully transparent, culled, or faded out by a **CanvasGroup** is skipped.

> **Include containers:** also picks up invisible panels, layout groups and the Canvas root, which is what you want when you are laying out a hierarchy rather than editing what is on screen.

> **Off:** UI is ignored by the overlay entirely.

Highlighting & feedback

Objects highlight as you move onto them, so it is always clear what a click will select. You can adjust how the highlight looks in the **Overlay Style** foldout of Settings.

You can tune the highlight's look in the Overlay Style settings.

Two overlay options are worth knowing about. The rest live in Settings.

> **Highlight before scroll:** when on, objects highlight as soon as you hover. When off, the highlight engages once you start scrolling to cycle. Pick whichever feels best.

> **Select Prefab Root:** hover or click a mesh inside a prefab and select the whole prefab instead of the single renderer, which is handy with deeply nested prefabs.

Tips

> **An object isn't registering?**
>
> Aim the cursor more toward the **mass of the geometry** rather than a thin edge, which usually resolves it. On very dense scenes, switching from Mesh to Collider detection can also speed things up. See Troubleshooting.
>
> If you're still seeing issues, perhaps check your Detection Layers in the Settings, and ensure the GameObject you're looking at is on an enabled Detection Layer.

The Context Menu

The Smart Context Menu is the second pillar of Smart Mouse. Press the **Context Menu key** (**Right Click** by default) anywhere in the Scene View and a menu of scene-building shortcuts opens at your cursor. Every action works at the spot you clicked, or on your current selection.

The Smart Context Menu, organised into Create / Selection / Measure / Windows.

> **Issue with Context Menu Keybind?**
> Depending on your Operating System, certain keybinds might not work as intended, for example, CMD + Space on MacOS. Most key combinations should work though, and I hope you'll find one that suits you :) Do feel free to contact me if this blocks you.

What's in the menu

The menu is grouped into **four categories**, so the most common actions are easy to find:

> **Create:** spawn empties, primitives, lights, particles, cameras and UI at the cursor, plus **Create ▸ Prefab** to drop your saved prefabs onto the surface below it.

> **Selection:** everything that acts on the selected objects, including Move to Mouse, Snap to Ground and Align & Snap, Settle With Physics, Group and Parent, Replace, and Variation.

> **Measure:** measure distances in the scene.

> **Windows:** quick access to Unity windows and your own project folders.

Items that act on a selection only appear when something is selected, so the menu stays as short as the moment needs.

Changing the activation key

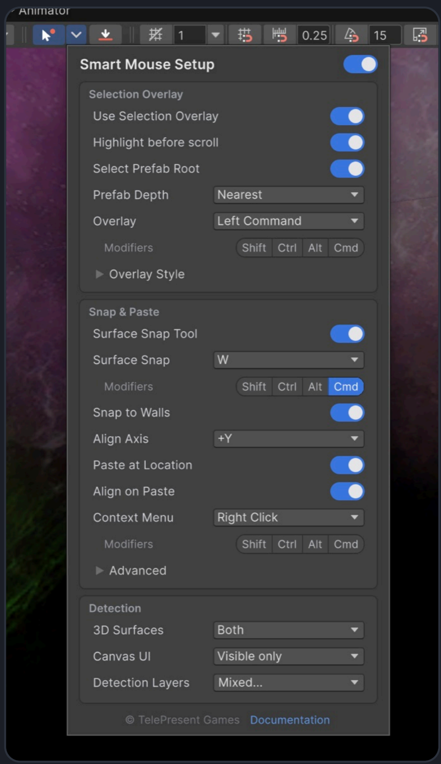
If right-click is inconvenient, or it clashes with another tool or input setup, reassign the **Context Menu Activation Key** to another mouse button or key in Smart Mouse Settings.

Make it your own

> **The menu is extendable**
>
> You can add your own entries with a single static method and one attribute, no editor boilerplate required. See Extending Smart Mouse.

Settings & Shortcuts

Press the small **dropdown arrow** next to the Smart Mouse toolbar button to open the settings popup. Everything is saved automatically, so your setup persists across projects and sessions.



The Smart Mouse Setup popup, with its Selection Overlay, Snap & Paste and Detection sections.

Enable Smart Mouse

The header of the popup (**Smart Mouse Setup**) carries the master **Enable Smart Mouse** switch, the same on/off as the toolbar button. While it is off, the rest of the settings are greyed out to show they aren't doing anything.

Selection Overlay

> **Use Selection Overlay:** master toggle for the hover highlight (on by default).

> **Highlight before scroll:** when on, objects highlight as soon as you hover. When off, highlighting engages once you scroll to cycle.

> **Select Prefab Root:** when on, hovering or clicking a mesh inside a prefab highlights and selects the whole **prefab** instead of the individual renderer. This also applies when clicking with the Surface Snap Tool. A **Prefab Depth** choice (Outermost or Nearest) then sets how far up nested prefabs it climbs.

> **Overlay:** the modifiers plus key you hold to engage the overlay (⌘ on macOS, Ctrl on Windows by default).

> **Overlay Style:** a foldout with **Show Hover Label** (the cursor label showing the object's name and world position) and **Ping on Hover** (flash the hovered object in the Hierarchy when the highlight changes), plus the highlight's fill and outline colours, outline width, glow and opacity.

Snap & Paste

> **Surface Snap Tool:** show or hide the Surface Snap button on the Smart Mouse toolbar. Click that button to toggle Surface Snap mode.

> **Surface Snap:** a shortcut (a key plus modifiers) that toggles Surface Snap mode. Defaults to ⌘+W on macOS and Ctrl+W on Windows; set the key to None to disable it.

> **Snap to Walls & Ceilings:** let surfaces that do not face up count as placement targets, so dragging or pasting onto a wall, an overhang or a ceiling mounts the object flush against it. With it off, only upward-facing surfaces are used for placement. On by default. See Snapping to walls and ceilings.

> **Align Axis:** which of an object's own local axes meets the surface when it aligns, from **+X** to **-Z**. The default **+Y** stands objects upright; **+Z** suits props that face out of a wall. Overhead the axis decides which way up a prop hangs: **+Y** turns it upside down under the ceiling, **-Y** keeps it the right way up. The same choice is on the Surface Snap window's split button. See Align Axis.

> **Paste at Location:** enable pasting at the cursor. When on, an **Align on Paste** option appears that aligns pasted objects to the surface normal.

> **Context Menu:** the modifiers plus key or mouse button that open the context menu (Right Click by default). With a modifier set, a plain right-click is left to Unity.

> **Advanced:** a foldout with two tabs of placement tuning, each with a live diagram. **Surface Sampling** holds **Ray Grid** (how many rays sample the surface), **Ray Spread** (how wide they sample), **Search Depth** (how far down they look) and the **Drag Re-sample** slider described on the Surface Snap page. **Alignment** holds **Min Slope to Align** (surfaces gentler than this stay flat), **Max Tilt** (the object never leans more than this) and **Surface Offset** (lift the object off the surface, or sink it in).

Detection

These options control *what* Smart Mouse detects under the cursor. They apply everywhere it reads the scene: the selection overlay, snapping and paste.

> **3D Surfaces:** which 3D surfaces to detect - **Meshes** (precise per-triangle hovering that works without colliders), **Colliders** (2D and 3D physics colliders, lighter on heavy scenes), **Both**, or **Off** (leaving only Canvas UI, if enabled).

> **Canvas UI:** detect world-space Canvas UI in the overlay. **Visible only** picks elements that actually draw; **Include containers** also picks up invisible panels, layout groups and the Canvas root; **Off** ignores UI.

> **Detection Layers:** choose which layers Smart Mouse detects. Unchecked layers are ignored everywhere (snapping, hover-selection and paste), for both their meshes and their colliders, so you can keep helper objects such as trigger volumes out of the way.

> **Ignore Raycast is excluded by default**
> In **Detection Layers**, the built-in **Ignore Raycast** layer starts *unchecked*, matching how Unity's own Scene view skips those objects when you click. Re-check it if you do want Smart Mouse to detect objects on that layer.

Keyboard & mouse reference

> **Hold ⌘ / Ctrl** (Overlay key): engage the selection overlay.

> **Mouse wheel** (while hovering): cycle through stacked objects.

> **Click / Shift-click**: select, or add and remove from the selection.

> **Right Click** (Context Menu key): open the Smart Context Menu.

> **⌘ / Ctrl + C, then ⌘ / Ctrl + V**: copy, then paste at the cursor (when Paste at Location is on).

> **Esc**: exit Measure mode.

The Surface Snap Tool's drag options (Keep Rotation, Center on Cursor, Maintain Offsets) live in its own in-scene window rather than on a modifier key. See its page for details.

> **All keys are re-bindable**
> Both the Overlay key and the Context Menu key can be reassigned to other keys or mouse buttons in this popup, which helps if a default clashes with your own setup.

Surface Snap Tool

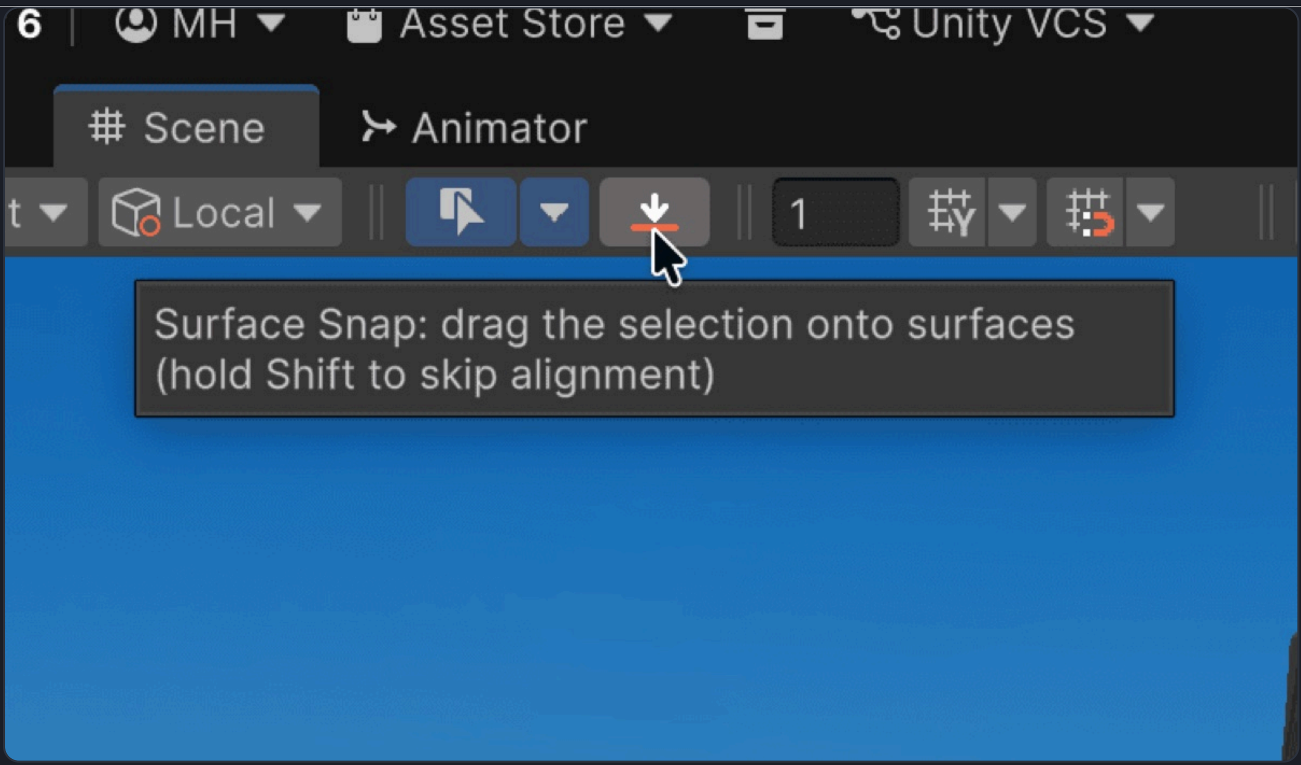
The Surface Snap Tool lets you **drag objects straight onto surfaces**. As you drag, the selection snaps down to the surface under the cursor and aligns to its normal, so placing a rock on a hillside or a sign on a slope is one smooth motion.

Dragging an object across the scene, snapping and aligning to the surface as it moves.

Finding & using the tool

Surface Snap is a button on the **Smart Mouse toolbar**, next to the enable button and the settings dropdown. Click it to toggle Surface Snap mode on or off; the button highlights while it is active. With it on, drag your selection across the scene. The regular transform gizmo is hidden while you snap.

You can also toggle the mode with a keyboard shortcut, ⌘+W on macOS and Ctrl+W on Windows by default. It can be rebound or disabled in the settings popup (the Surface Snap key).



The Surface Snap button on the Smart Mouse toolbar, highlighted while active.

Showing or hiding the button

Don't need it? You can remove the button from the toolbar, and bring it back, from two places. Both flip the same setting:

> The **Surface Snap Tool** toggle in the Smart Mouse Settings popup.

> **Tools ▸ TelePresent ▸ Smart Mouse ▸ Show Surface Snap Tool**.

The button is shown by default. Hiding it while it is active also switches the mode off.

Using it

> **Drag** the selected object across the scene. It follows the cursor, sits on the surface beneath it, and rotates to match the surface normal.

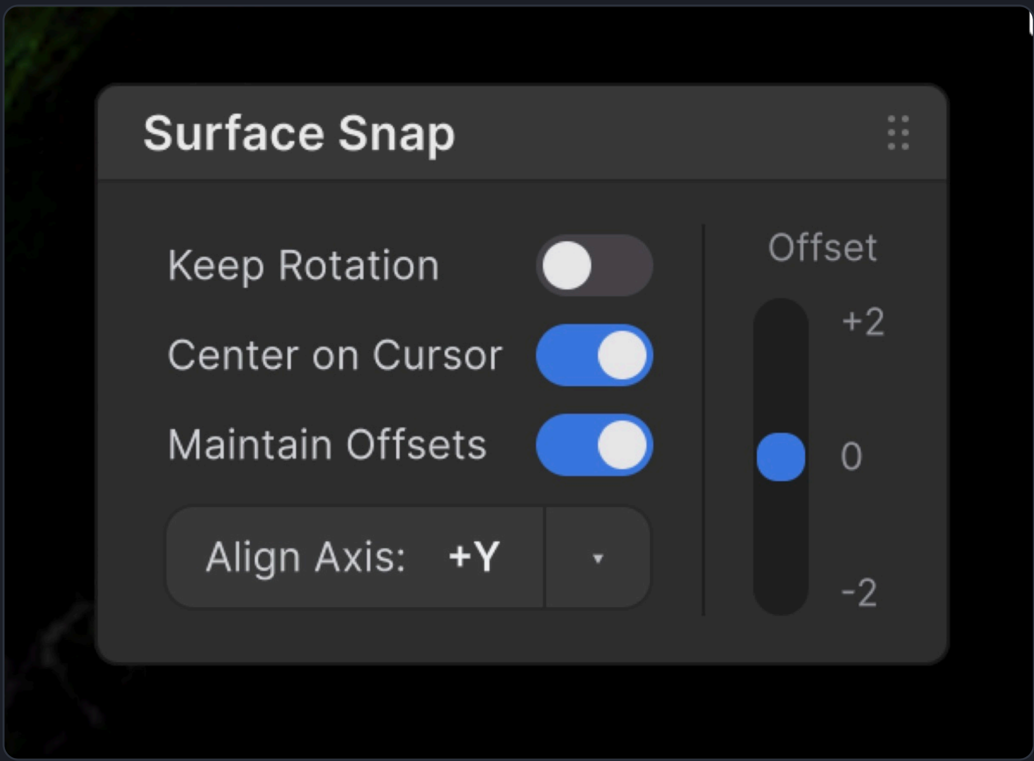
> **Click without dragging** to select a different object; the tool doesn't trap your selection. Clicking respects your Select Prefab Root setting, so you can grab the whole prefab rather than the individual mesh you clicked.

> **Undo** reverts the entire drag in a single step.

Three drag options live in the tool's own settings window, described below: Keep Rotation, Center on Cursor and Maintain Offsets.

Drag options

While Surface Snap is active, a small **Surface Snap** window appears in the corner of the Scene View. Its toggles are remembered between sessions, and the window disappears when you switch the tool off.



Surface Snap Mode Window with Alignment Axis Setting

> **Keep Rotation:** snaps position only. Each object keeps its current rotation instead of aligning to the surface normal. Leave it off for the usual hug-the-surface behaviour.

> **Center on Cursor:** moves the selection so the cursor becomes the centre of the group. With several objects selected they keep their relative layout, but the whole arrangement is centred under the pointer instead of grabbed from the spot you first clicked.

> **Maintain Offsets:** snaps the selection as one rigid group. Every object keeps its relative position and rotation, so a stacked or carefully arranged set stays assembled, and each piece keeps its original height off the surface instead of dropping flush onto the ground.

Align Axis

Align Axis decides which of an object's own local axes meets the surface. The default is **+Y**, so the object's up axis leaves the surface and it stands upright on whatever it lands on. Change it when you wish for an Object to align differently, for example, against a wall.

Overhead the axis is the orientation control. Whichever axis you pick is the one that turns to meet the surface, so under a ceiling the default **+Y** hangs a prop upside down and **-Y** keeps it the right way up.

The control appears twice. In the **Surface Snap** window it is a split button: the main part reads **Align Axis:** followed by the current value, and the **dropdown** opens the full list. The same choice is in the settings popup under **Snap & Paste**. Both read the same saved value, but changing it from the Surface Snap window also re-aligns the current selection straight away, while the settings popup only stores it for the next placement.

> **Click the button** to cycle the letter X, Y, Z while keeping the sign, so +Y becomes +Z and -X becomes -Y.

> **Click the dropdown** to pick any of the six values from a menu, with a tick on the current one.

Snapping to walls and ceilings

By default the tool assumes you are placing things on the ground, and drops them straight down. **Snap to Walls & Ceilings**, in the settings popup under **Snap & Paste**, lets surfaces that do not face up count too, so you can drag a prop onto a wall, an overhang or a ceiling and have it sit flush against it. It is on by default.

One rule covers all of them. A surface that faces up enough is ground, and the object is set down on it. Anything steeper is mounted flush along the surface normal instead. Because there is only that one boundary, a drag that sweeps from a wall up over a curved roof to fully overhead never changes behaviour halfway.

With the setting off you get the old ground-only behaviour back: only upward-facing surfaces are used for placement, and walls, overhangs and ceilings stop counting as targets.

It works for pasting too. See Paste at Hover Location.

> **Mounted placement skips the sampling settings**
> The **Surface Sampling** options, plus **Min Slope to Align** and **Max Tilt**, only shape ground placement. On a wall or a ceiling the object follows the surface normal directly, so those sliders have no effect there.

Dragging from the Project window

If you have Surface Snap Mode enabled while dragging from the Project Window, the Project Window Object will automatically Snap and Align with the surfaces, just like when you use Surface Snap Mode normally!

Hold **Shift** as you drop to place the prefab unrotated. Dragging from the Project window needs Unity 2022.1 or newer. On older editors the drag falls back to Unity's own behaviour.

Dragging multiple objects

With several objects selected, the tool keeps their relative layout while conforming *each* one to the surface beneath it. A scattered cluster of props stays arranged as you laid it out, but every piece settles onto the terrain individually, which is ideal for moving a group of rocks or foliage onto uneven ground. Turn on **Center on Cursor** (above) if you would rather the whole group be centred on the pointer as you drag.

> **Tip**
> On busy, multi-layered terrain, dragging objects one or a few at a time gives the most predictable result, since the surface directly under each object is what it snaps to.

> **Dragging a big selection?**
> The **Drag Re-sample** slider (in the settings popup under **Advanced ▸ Surface Sampling**) controls how far an object moves before its surface tilt is re-sampled. The default of 0 re-samples every frame for the smoothest result; raising it makes large drags noticeably faster, at the cost of the tilt updating a little behind on uneven ground. It applies to both individual drags and **Maintain Offsets** group drags.

Dragging Canvas UI

Select a UI element and its canvas is outlined, showing the plane the drag will happen on. Dragging slides the selection along that canvas, keeping each element's depth.

The canvas is taken from the element's nearest parent **Canvas**, so an element inside a nested canvas moves along that one. All render modes work, including Screen Space.

Two cases are refused, with a warning in the console:

> **A parent layout group** owns its children's positions, so anything you drag would be put back on the next layout pass.

> **A Screen Space canvas root** is positioned by Unity every frame. Drag the elements inside it instead.

PLACING ON SURFACES

Paste at Hover Location

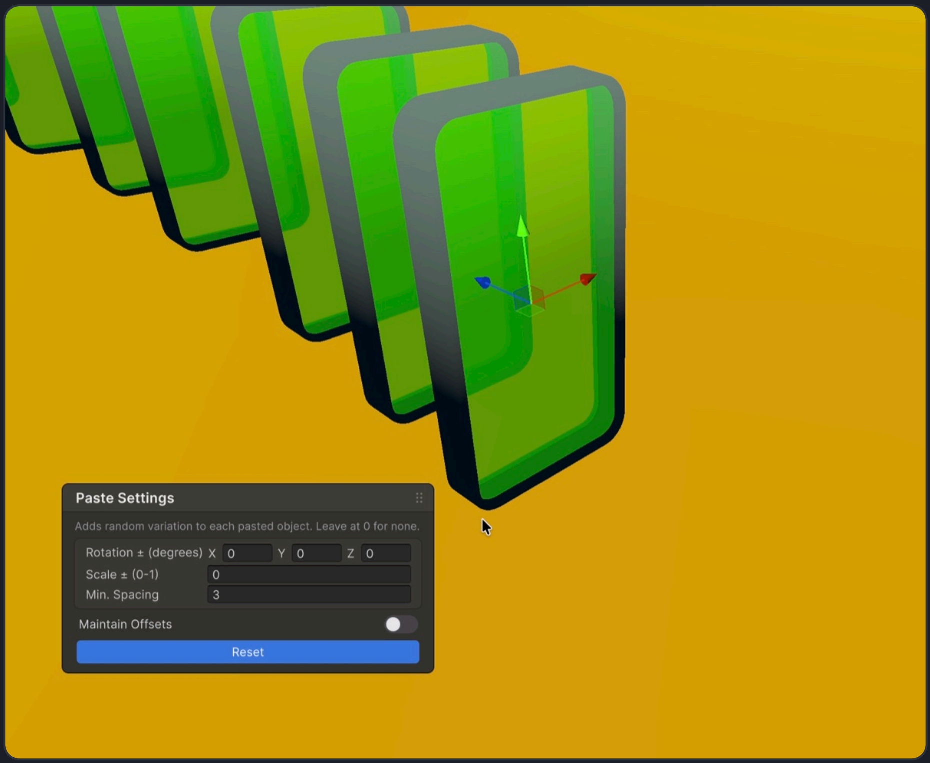
With **Paste at Location** enabled in Settings, the usual **⌘/Ctrl + V** is overridden inside the Scene View. Smart Mouse casts a ray to the geometry under your cursor and pastes the copied objects at that exact hit point, instead of back at their original position.

Copy as you normally would (**⌘/Ctrl + C**), from either the Scene View or the Hierarchy, then move the cursor over a surface and paste.

Paste at Location drops each copy exactly where the cursor points.

Align on Paste

With **Align on Paste** turned on, pasted objects are oriented to match the surface normal they land on, so they sit flush against slopes and walls instead of staying axis-aligned.



With Align on Paste on, pasted objects orient to the surface they land on.

Matching the original's alignment

With **Align on Paste** on, a copy does not simply adopt the Align Axis. Smart Mouse first looks at what the original was resting on and reproduces that same relationship on the new surface, so a sign copied off a wall arrives facing out of the wall you paste it onto, a lamp copied off a ceiling keeps hanging under the ceiling you paste it onto, and a rock copied off the ground arrives sitting on the ground.

> The **Align Axis** is the fallback. It is used when a copy cannot be matched back to its original, or when that original was not resting on anything.

> With **Align on Paste** off, nothing is oriented at all and the axis is never read.

Pasting onto walls and ceilings

With **Snap to Walls & Ceilings** on in Settings, pasting works on any surface that does not face up as well as on the ground. Aim at a wall or a ceiling and the copies mount flush against it instead of dropping to the floor below.

> **A wall copy pasted onto the floor lies down**
>
> Matching the original's relationship is what makes a wall copy stay on a wall, and it applies in the other direction too. Paste a wall-mounted sign onto the floor and it arrives face down, because that is the same relationship reproduced on a flat surface. Turn **Align on Paste** off, or nudge it afterwards, if you want it standing.

> **Pasting overhead needs the setting on**
>
> With **Snap to Walls & Ceilings** off, a paste aimed at a ceiling has nothing to fall to and no existing position to hold, so the copy lands inside the slab. Turn the setting on to paste overhead.

The Contextual Pasting Menu

When you paste, a small window appears in the corner of the Scene View. It lets you add controlled randomness so repeated pastes look natural rather than identical:

> **Rotation ± (degrees):** random rotation (per axis) applied to each pasted object.

> **Scale ± (0-1):** random uniform scale variation, so each pasted object keeps its proportions while its overall size varies.

> **Min. Spacing:** the minimum distance the cursor must move before another paste is allowed. This is perfect for holding ⌘/Ctrl+V to scatter objects continuously without them clipping into each other.

> **Maintain Offsets:** pastes the selection as one rigid group. Every object keeps its relative position and rotation (a stacked arrangement stays assembled) and its height off the surface, instead of each object dropping flush to the ground.

The window dismisses itself as soon as you give any unrelated input.

> **Great for set dressing**
>
> A little random rotation and scale goes a long way toward making repeated props such as rocks, plants and debris read as a natural, hand-placed scene.

Placing prefabs and existing objects

To drop your own prefabs at the cursor, see Create ▸ Prefab. For placing objects that are already in the scene, see the Surface Snap Tool and Snap & Align.

Pasting onto a canvas

With a UI element selected, pasting places the copy on that element's canvas at the cursor, keeping its depth and its original parent. It works over empty canvas space as well as over other elements, because the canvas plane is used when nothing is directly under the cursor.

Pasted UI is never dropped onto a surface or tilted to a normal, which would take it off its canvas.

Snap & Align

These context-menu actions reposition objects that are already in the scene. Select one or more objects, right-click, and choose an action from the **Selection** category. Each is a single, undoable step.

Snap to Ground

Drops each selected object straight down so its **lowest point** rests on the first surface beneath it. Rotation is left untouched, which is handy for settling objects onto a floor or terrain without changing their facing.

Align & Snap

Like Snap to Ground, but also **rotates each object to match the surface normal** it lands on, so objects sit flush against slopes and angled surfaces.

> **How the surface is found**
>
> Each object looks for a surface *below* it first, then *upward* if it doesn't find one, which explains the occasional surprise on stacked or layered terrain. It works with or without colliders, so a mesh surface alone is enough.

Move to Mouse

Under **Selection**, this moves the whole selection to the cursor's position in the scene, keeping the objects' relative arrangement and dropping them onto the surface there. It is the one-shot counterpart to the interactive Surface Snap Tool, so use whichever suits the moment.

Create Objects

The **Create** category in the context menu spawns new objects at the cursor. The object lands on the surface under the cursor (or a short distance along the cursor ray if you are pointing at empty space) and is selected, ready to tweak.

What you can create

Empty GameObject

2D Object: Sprite

3D Object: Cube, Sphere, Capsule, Cylinder, Plane, Quad

Effects: Particle System

Light: Directional, Point, Spot, Area

Audio: Audio Source

UI: Canvas, Button

Camera

Volume: Box Volume, Sphere Volume

The Create submenu, for spawning objects at the cursor.

Your own prefabs

Create ▸ Prefab holds a list of your saved prefabs. Pick one and it drops at the cursor, on the surface beneath it and aligned to the normal when Align on Paste is on. The new instance stays connected to its prefab, and placing it is a single undo.

To build that list, choose **Create ▸ Prefab ▸ Manage Prefabs…** (also at **Tools ▸ TelePresent ▸ Smart Mouse ▸ Manage Prefabs…**). In the window that opens:

> **Add New Prefab Slot** adds an empty slot.

> Drag a prefab into a slot's object field.

> **Remove** takes a prefab back off the list.

> **Save and Close** stores your list.

Your prefabs then appear under Create ▸ Prefab, ready to place at the cursor. Until you add one, the submenu shows *No prefabs added*.

Every created object is registered with **Undo**, so a single Ctrl/⌘+Z removes it. To place objects that are already in the scene, see Paste at Hover Location and the Surface Snap Tool.

Variate Transforms

Variate Transforms applies **randomised rotation and scale** to the selected objects. It is a fast way to break up the repetition of duplicated props and make a scene feel hand-placed. Find it under **Selection ▸ Variate Transforms** in the context menu.

Quick Variation

Opening it shows a small window in the Scene View with a slider per rotation axis (X / Y / Z) and a **Scale ±** slider. As you drag a slider, each selected object is given a random value within ± that range, updating live so you can dial in the look. A common recipe is a generous **Y** range (spin) with little or no X/Z.

Scale is locked across X, Y and Z, so every object keeps its proportions while its overall size varies. Use it for fields of rocks, foliage or debris where one repeated size would look artificial. The scale range starts at **0** (no scale variation), so out of the box the window only varies rotation; raise the slider when you want size variety too.

The two channels are independent: dragging a rotation slider re-rolls only the rotations, and the scale slider only the scales, each from the values the objects had when the window opened. That way you can fine-tune one without disturbing a result you already like on the other.

The Variate Transforms window, with rotation sliders per axis and a scale range slider.

> **Live preview & undo**
>
> The window previews its effect on the current selection as you adjust the sliders, and the result is captured as a single undo step. **Reset** returns the ranges to their defaults.

Want variety *as you paste* rather than after? The Contextual Pasting Menu applies the same kind of random rotation and scale to each object as it's dropped.

Grouping & Parenting

Quick ways to organise a selection's hierarchy, found under **Selection** in the context menu (they appear when more than one object is selected).

Group Selected Objects

Creates a new empty **Group** GameObject at the **average position** of the selection and re-parents every selected object under it. A tidy way to bundle related objects without manually creating a parent and dragging things in. With objects from several loaded scenes selected, each scene gets its own group, so nothing is moved between scenes.

Parent to Last

Parents every selected object to the **last-selected** object. World positions and sizes are preserved. Objects in a different scene than the parent, or ones that would create a loop, are skipped with a console warning.

Replace Objects

Replace Selected Objects swaps every object in the current selection for a prefab of your choosing, in place. It is ideal for blocking out a scene with placeholders, then swapping them all for final art in one move. Find it under **Selection** in the context menu.

Using it

> Select the objects you want to replace, then choose **Selection ▸ Replace Selected Objects**. A small window opens in the Scene View.

> Drag your **Replacement Object** (a prefab) into the field.

> Choose whether to **Keep Scale** and **Keep Rotation** from each original object, then press **Replace Selected Objects**.

Each original is removed and a new instance of the prefab takes its place, keeping the original's parent and order in the Hierarchy (and, if enabled, its scale and rotation). The new objects become the selection, and the whole replacement reverts in a single undo step. Children inside a prefab instance, and the root of a prefab open in Prefab Mode, can't be replaced on their own; those are skipped with a console warning.

The Replace Selected Objects window, with its Replacement Object field and Keep Scale / Keep Rotation toggles.

Measure Tool

The Measure tool reports the **world-space distance** between two points in your scene. It is handy for checking gaps, sizing rooms or spacing props. Enable it from **Measure ▸ Enable Measurement Mode** in the context menu, or from **Tools ▸ TelePresent ▸ Smart Mouse ▸ Toggle Measurement Mode**.

Measuring

> **Click and drag** across a surface to set point **A** (start) and point **B** (end). A line connects them and the distance is shown both at the line and in the status bar.

> **Release** to lock the measurement in place.

> **Click once** (without dragging) to clear the current points.

> **Press Esc** to exit measurement mode.

A measurement between points A and B, with the connecting line and a distance readout.

> **Note**
>
> Measurement points snap to the surface under the cursor using the same detection as the selection overlay, so Smart Mouse must be enabled while you measure.

Settle With Physics

Settle With Physics drops props and lets them fall and settle onto your geometry using physics. It happens in Edit Mode, right in the Scene View, and leaves no temporary components behind.

How to use it

Select one or more objects in the scene.

Right-click in the Scene View and choose **Selection ▸ Settle With Physics**.

The objects **drop and settle** onto whatever is beneath them, then come to rest.

Press the on-screen **Stop Physics** button to end a run early.

A single **Undo** (⌘/Ctrl + Z) returns everything to exactly where it started.

Dropping the selection and letting it settle onto the scene with physics.

Quick Windows & Folders

The **Windows** category in the context menu gives you keyboard-free access to common editor windows and your own most-used project folders, without leaving the Scene View.

Unity windows

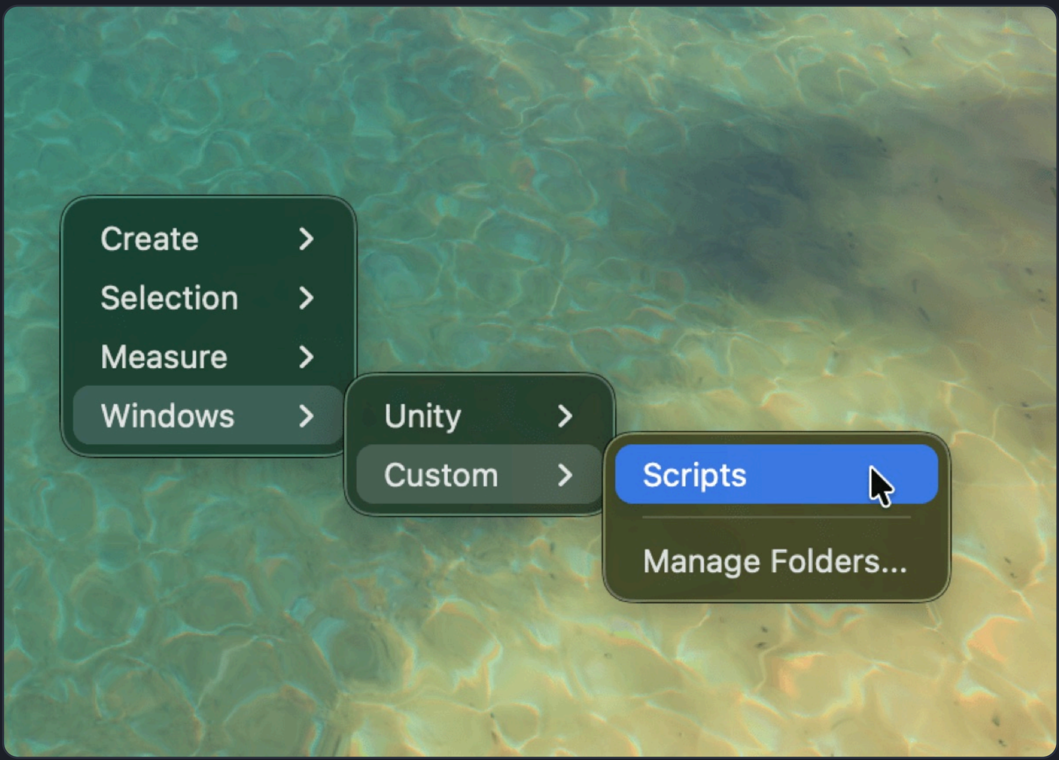
Open frequently used editor windows directly from the menu, including:

> Lighting, Inspector, Hierarchy, Project, Console

> Game, Animator, Animation, Profiler

Custom folder shortcuts

Add shortcuts to the project folders you open most. Choose **Windows ▸ Custom ▸ Manage Folders…** to open the manager, then add folder paths (type a path like `Assets/Art/Props`, or use **Browse** to pick one). Each saved folder appears in the menu; selecting it pings and focuses that folder in the Project window.



The Manage Folders window, listing saved project-folder shortcuts.

> **Also in the Tools menu**
> The prefab and folder managers can also be opened from **Tools ▸ TelePresent ▸ Smart Mouse**.

Extending Smart Mouse

The Smart Context Menu is fully extendable. You can add your own entries by writing a single **static method** and marking it with one attribute. Use them to run a custom tool, kick off a build step, spawn a special prefab, or anything else. No editor windows or boilerplate required.

Your first menu item in two short lines

Give the attribute a menu path and put it on a **parameterless static method**. Smart Mouse registers the item for you, and the method runs when it's clicked:

```
using TelePresent.SmartMouse;
using UnityEngine;

public static class MyMenuItems
{
    [SmartMouseContextMenuItem("Example/Call Method")]
    static void CallMethod() => Debug.Log("Hello from the Smart Mouse context menu!");
}
```

The path defines the **category** and the **item name**, split by `/`. If your action works on the selected objects, add `RequiresSelection = true` and Smart Mouse greys the item out automatically while nothing is selected:

```
[SmartMouseContextMenuItem("Example/Drop To Ground", RequiresSelection = true)]
static void DropToGround()
{
    foreach (GameObject go in UnityEditor.Selection.gameObjects)
    {
        UnityEditor.Undo.RegisterCompleteObjectUndo(go.transform, "Drop To Ground");
        Vector3 p = go.transform.position;
        go.transform.position = new Vector3(p.x, 0f, p.z);
    }
}
```

To try it: save the script as `MyMenuItems.cs` inside a folder named **Editor** (create one anywhere in your project if you don't have one), let Unity compile, then right-click in the Scene View. Your *Example* category appears after the built-in ones. If you keep your editor scripts in their own **assembly definition**, also add a reference to `TelePresent.SmartMouse` in that asmdef so it can see the attribute.

That's the whole setup for simple uses. When you need more, such as several items from one method, separators, or the cursor's position in the world, use the full form below.

Building richer menus

For more control, mark a static method with `[SmartMouseContextMenuItem]` and *no path*. Smart Mouse calls it whenever the menu is built, passing you the `GenericMenu` to add to, plus the current `SceneView`, the cursor's screen position, and a world-space `Ray` from the cursor:

```
using UnityEditor;
using UnityEngine;
using TelePresent.SmartMouse;

public static class MyMenuItems
{
    [SmartMouseContextMenuItem]
    public static void Register(GenericMenu menu, SceneView sceneView,
                                Vector2 mousePosition, Ray worldRay)
    {
        // category "Example", item "Call Method", not checked, runs CallMethod
        menu.AddItem(new GUIContent("Example/Call Method"), false, CallMethod);
    }

    private static void CallMethod()
    {
        Debug.Log("Hello from the Smart Mouse context menu!");
    }
}
```

The `GUIContent` path works exactly like the simple form's. The `false` is the checkmark state, and the last argument is the method to run when the item is clicked. Because your method receives the menu itself, it can add as many items, separators, and disabled entries as it likes.

Acting on the selection

The two-line form already covers most selection actions, as the *Drop To Ground* example above shows. The full form adds a second moment to work with the selection: while *building* the menu, to decide what to show. Capture the selection in a local variable and the callback receives exactly what was selected when the menu opened:

```
[SmartMouseContextMenuItem]
public static void Register(GenericMenu menu, SceneView sceneView,
                            Vector2 mousePosition, Ray worldRay)
{
    GameObject[] selected = Selection.gameObjects;
    if (selected.Length == 0) return; // nothing selected: add no items

    menu.AddItem(new GUIContent("Example/Snap Selection To Grid"), false, () =>
    {
        foreach (GameObject go in selected)
        {
            Undo.RegisterCompleteObjectUndo(go.transform, "Snap To Grid");
            Vector3 p = go.transform.position;
            go.transform.position = new Vector3(Mathf.Round(p.x), Mathf.Round(p.y),
Mathf.Round(p.z));
        }
    });
}
```

The menu is modal, so the selection cannot change between the menu opening and the click; reading `Selection.gameObjects` inside the callback would give the same objects. Capturing it in a local, as above, just makes that explicit.

> **Make it undoable**
>
> If your action changes objects, register the change with Unity's Undo system first, as the example does: `Undo.RegisterCompleteObjectUndo` before modifying an existing object, or `Undo.RegisterCreatedObjectUndo` after creating a new one. Users expect Ctrl+Z to work on everything in the menu, and every built-in Smart Mouse action supports it.

Placing objects at the cursor

The `worldRay` parameter is a world-space ray from the camera through the cursor, captured when the menu opened. To find the surface it points at, use `SmartMouseRaycast`: your item gets the same detection the tool itself uses, meshes with no colliders included:

```
[SmartMouseContextMenuItem]
public static void Register(GenericMenu menu, SceneView sceneView,
                            Vector2 mousePosition, Ray worldRay)
{
    menu.AddItem(new GUIContent("Example/Spawn Marker Here"), false, () =>
    {
        // The surface under the cursor, or 5 units along the ray when nothing is hit.
        Vector3 point = SmartMouseRaycast.GetPointOrDefault(worldRay, 5f);

        GameObject marker = GameObject.CreatePrimitive(PrimitiveType.Sphere);
        marker.name = "Marker";
        marker.transform.position = point;
        Undo.RegisterCreatedObjectUndo(marker, "Spawn Marker");
        Selection.activeGameObject = marker;
    });
}
```

When you also want the surface normal, or the object that was hit, use the full form. Here the marker is aligned to the surface it lands on:

```
if (SmartMouseRaycast.TryGetSurfaceHit(worldRay, out Vector3 point,
        out Vector3 normal, out GameObject surface))
{
    marker.transform.SetPositionAndRotation(point,
        Quaternion.FromToRotation(Vector3.up, normal));
}
```

The normal faces the viewer rather than following the mesh winding, so two-sided and back-facing geometry still align correctly. Versions before 2.2 returned the raw winding normal.

A plain `Physics.Raycast` works too when your scene has colliders everywhere. If your editor scripts live in their own assembly definition, reference `TelePresent.SmartMouse.Editor` for `SmartMouseRaycast` (the attribute alone comes from `TelePresent.SmartMouse`).

Ordering your items

Pass an optional number to the attribute to control where your category lands relative to others: `[SmartMouseContextMenuItem(50)]`, or with a path, `[SmartMouseContextMenuItem("Example/Do Thing", 50)]`. Lower numbers appear earlier. Smart Mouse's own groups use 10–40, so without a number your items default to a high value and list neatly **after** the built-in ones.

Conditional items

Because your method runs each time the menu opens, you decide what to show on the spot: guard the `AddItem` call, or use a **disabled item** to tell the user why an action isn't available right now, exactly like the built-in entries do:

```
[SmartMouseContextMenuItem]
public static void Register(GenericMenu menu, SceneView sceneView,
                            Vector2 mousePosition, Ray worldRay)
{
    if (Selection.gameObjects.Length >= 2)
        menu.AddItem(new GUIContent("Example/Link Objects"), false, LinkObjects);
    else
        menu.AddDisabledItem(new GUIContent("Example/Link Objects (select two or more)"));
}
```

`menu.AddSeparator("Example/")` draws a divider line inside your submenu, which helps once it grows past a few entries.

Editor-only

Context-menu methods are editor code. Keep them in an **Editor** folder (or guard them with `#if UNITY_EDITOR`) so they don't ship in builds. If you get stuck wiring one up, send me an email and I'm happy to help.

Exceptions are contained

If your method throws, Smart Mouse logs the full exception to the Console with a clickable stack trace pointing at the failing line, and the rest of the context menu still opens. A bug in one extension never takes the menu down.

Item not showing up?

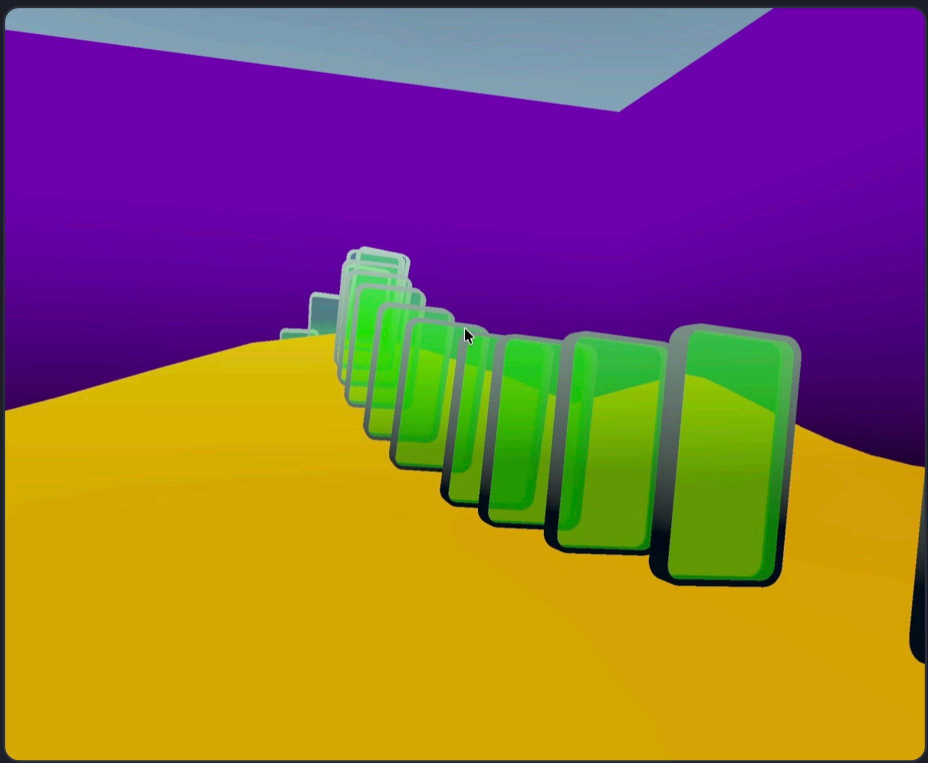
Check three things. The method must be **static** and match one of the two shapes: a path on the attribute with **no parameters**, or no path with exactly the four parameters `(GenericMenu, SceneView, Vector2, Ray)`. A method with the attribute but the wrong shape is skipped, with a Console error naming it. The script must compile, so clear any errors in the Console. And Smart Mouse itself must be enabled, since the context menu only opens while it is on.

Demo Scene

The fastest way to get a feel for Smart Mouse is to play with the included demo. It ships with a small scene and a few props (a desert ground plane and some glass panes) that are good for trying out hovering, pasting and surface snapping.

Opening the demo

In the Project window, open the **SmartMouseDemo** scene from **Assets/TelePresent/Smart Mouse**. Make sure the Smart Mouse toolbar button is enabled (it lights up blue). See Installation if you don't see it.



The included Smart Mouse demo scene, a small sandbox for hovering, pasting and surface snapping.

Things to try

> **Hover & select:** hold the Overlay Key (⌘ on macOS, Ctrl on Windows) and move the cursor over the props to highlight them. Scroll the mouse wheel to cycle through overlapping objects, then click to select.

> **Paste on a surface:** select a prop, copy it (⌘/Ctrl+C), then with Paste at Location enabled, ⌘/Ctrl+V drops a copy exactly where you are pointing, aligned to the slope of the dune.

> **Surface Snap drag:** pick the Surface Snap Tool and drag a prop across the terrain. It hugs and aligns to the surface as you move. Try it with several objects selected.

> **The context menu:** right-click in the viewport to open the Smart Context Menu and create, group, replace or measure things.

Troubleshooting

Common questions and quick fixes. If something here doesn't cover your issue, please reach out on the Discord or by email, and I'm always happy to help.

I don't see the Smart Mouse toolbar button

> After importing, the button appears on the **Scene View toolbar**. If it's missing, try **restarting the editor** first.

> Otherwise, open the Scene View's **overlay menu** (the three-dots/overlays button in the top-right of the Scene View), enable **Smart Mouse Toolbar**, and drag it up into the toolbar to dock it.

> You can also reach the main actions from **Tools ▸ TelePresent ▸ Smart Mouse**.

The selection overlay is slow to initiate

On complex scenes with a lot of mesh data to parse, the overlay can take a moment to build its first time after you start hovering. 2.0 includes a significant performance pass that makes this much faster, but on very heavy scenes you can still speed things up:

> In Smart Mouse Settings, set **3D Surfaces** to **Colliders** instead of Meshes. That skips the per-triangle mesh work entirely.

My objects paste above or below the surface I wanted

> **How snapping searches for a surface**
>
> When pasting, each object tries to snap to the surface found *below* it. If none is found, the search continues *upward* until a surface is hit.

> **Pasting too high:** make sure your surface has an active mesh, skinned mesh or terrain. If it does and you still see issues, try adding a **collider** to the surface.

> **Pasting too low:** the object found a viable surface *beneath* the one you intended (common on uneven terrain with overlapping surfaces). Paste **fewer objects at once** in those situations, or use the Surface Snap Tool to place them one drag at a time.

> **Floating above a scaled primitive:** Unity's sphere and capsule colliders stay round however you scale the object, so a flattened sphere keeps a round collider and the object snaps to that instead of the mesh. Set **3D Surfaces** to **Meshes** under Detection, or give the object a Mesh Collider.

The selection won't register an object

> Aim the cursor more toward the **mass of the geometry** rather than a thin edge, which usually resolves it.

> Check that the matching detection type is enabled in Settings (3D Surfaces and Canvas UI).

> The overlay highlight itself ignores scene **gizmos** (such as light icons); those aren't selectable through it.

The context menu doesn't open on right-click

> Make sure Smart Mouse is **enabled** (the toolbar button is blue).

> If right-click is being intercepted (for example by another tool or input setup), re-assign the **Context Menu Activation Key** to another key or mouse button in Settings.

Settle With Physics does nothing

> Settle With Physics uses 3D physics, so the selection needs 3D content (a mesh, or an existing Rigidbody/Collider) and a surface below it to land on. Any collider in the scene works as the ground. It's built in now, with no package to install.

If anything is getting in your way, please contact me and I'll do my best to help. 🙂

Thank You for Using Smart Mouse!

Smart Mouse 2.0 is a big labour of love, and I hope you'll find great enjoyment in it, as you work in Unity. I would love to keep supporting the tool as much as I can, so please do feel free to reach out to me with feedback anytime, whether it be on email, or in the Discord

This asset keeps evolving, and your feedback shapes where it goes. If you hit a problem, have a feature request, or just want to show what you have made, reach out at TelePresentGames@gmail.com or join the Discord.

Thanks again. 🙂

Happy developing!

Kind regards,
Martin

Changelog

2.2

The update is a big one! :) Improved smoothness, wall + ceiling support and canvas stuff, + using Surface Snap Mode to Drag Objects straight from the Project Window!

- **New: drag prefabs straight from the Project window.** With Surface Snap active, a prefab dragged out of the Project window follows the surfaces under your cursor and drops aligned. Hold **Shift** to drop it unrotated. Needs Unity 2022.1 or newer.
- **Snap to Walls & Ceilings.** Drag or paste a prop onto a wall or under a ceiling and it mounts flush. If you prefer everything staying grounded, the switch is in the settings popup, under Snap & Paste.
- **Better pasting.** Copies keep their original relationship to the surface they were resting on.

2.1

Thanks to everyone who provided feedback and suggestions! :)

Here's a bigger feature and polish update! The workflow around snapping to walls and working with Canvas UI has been majorly improved + Alignment settings have been improved :)

- **New: Canvas UI.** Canvas elements can now be easily dragged around the canvas, directly with Surface Snap Mode.
- **New: Snap to Walls.** Near-vertical surfaces now count as placement targets, so you can drag or paste a prop onto a wall and have it sit flush against it.
- **New: Align Axis.** Choose which of an object's local axes meets the surface, from the Surface Snap window or the settings popup. +Z suits props that face out of a wall.
- **Paste at Location** now reproduces each copy's original relationship to the surface it was resting on, instead of forcing one axis.
- Double clicking the Offset Setting in the Surface Snap Mode now resets it to 0.
- Various placement and stability fixes across snapping, pasting and the selection overlay.

2.02

- Fixed console warnings while holding the selection key in Linear color space projects.
- Improved Support for 2D Projects.

2.01

- Unity 6.5 Compatibility Patch.

2.0

A huge update, including many new features, optimizations, and smoother Unity integration

1.2

- **New macro:** move selected objects to the mouse position.
- Minor performance tweaks.

1.1

- Unity 2021 support. If you have trouble with the context menu, reassign the context-menu key away from the right mouse button in Smart Mouse Settings.
- Additional keybinds for the context menu are now supported.

Original release

- The first public release of Smart Mouse.

> **Keeping up to date**
>
> New versions are announced on the Discord or in the built-in Welcome Window. Have an idea or a bug to report? then please feel free to reach out anytime :)